

VIGNESH KUMAR EKAMBARAM

TECH LEAD | FULL-STACK SOLUTION ARCHITECT | CLIENT DELIVERY LEAD

Chennai, Tamil Nadu, India | 12+ years in software development | Open to international and on-site assignments

PROFESSIONAL PROFILE

Tech Lead and Full-Stack Solution Architect with 12+ years of experience delivering distributed, high-performance enterprise software. Leads a five-member engineering team, manages client priorities and delivery coordination, and combines hands-on expertise in .NET, Angular, SQL Server, AWS, automation, machine learning, and AI/RAG initiatives. Has supported client engagements through multiple on-site visits to Thailand and a client visit to Japan.

LEADERSHIP & DELIVERY STRENGTHS

- **Team leadership:** Lead a five-member engineering team; plan work, unblock delivery, coordinate stakeholders, and manage client requests.
- **International client support:** Supported client visits and delivery activities in Thailand and Japan.
- **Quality engineering:** Improved browser-level regression coverage using Playwright automation and MCP-enabled testing workflows.
- **Cloud delivery:** Experience with AWS EC2, S3, ELB/load balancing, CloudSearch, WAF, and RDS for SQL Server and MySQL.

PROFESSIONAL EXPERIENCE

Tech Lead

alfaTKG
2022 - Present

- Lead the engineering delivery of JQMS, PTE, and AlfaDock using .NET, Angular, Node.js, and SQL Server.
- Manage a five-member team, prioritize client requirements, coordinate delivery, and provide direct client support.
- Designed asynchronous processing with RabbitMQ, AWS SQS, and .NET worker services; used SignalR/WebSockets for live feedback.
- Optimized high-concurrency SQL Server workloads through Query Store analysis, execution plans, locking analysis, and indexing.
- Architected AWS deployments and automated GitLab CI/CD pipelines for multi-tenant IIS environments.

Senior Software Engineer

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2018 - 2022

- Built machine-learning workflows for sheet-metal quotation and machine-cycle-time prediction.
- Applied HOG feature extraction, YOLO-based visual detection, and regression approaches including DecisionTreeRegressor to manufacturing data.
- Designed modular quotation engines for laser cutting, punching, bending, welding, cost estimation, and BOM generation.
- Developed core architecture for AlfaDock and PTE, and mentored engineers on testing and design practices.

Software Development Engineer

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2015 - 2018

- Developed .NET and SQL Server applications, REST APIs, optimized database access, and provided production incident support.

- Built backend APIs and relational database structures with C# and SQL Server across the full software-development lifecycle.

SELECTED ENGINEERING HIGHLIGHTS

Manufacturing quotation intelligence. Automated sheet-metal quotation workflows with CAD-driven inputs, nesting calculations, laser-cutting paths, bending operations, and machine-runtime estimation.

AI agents and RAG. Designed specialized quotation, production scheduling, and machine-data agents coordinated through an orchestrator using Qdrant retrieval and LLM reasoning.

Resilient enterprise systems. Implemented YARP API gateway patterns, JWT authentication, Redis caching, rate limiting, and decoupled worker-based processing.

TECHNICAL SKILLS

Languages & Frameworks C#, TypeScript, JavaScript, SQL, ASP.NET Core, Web API, EF Core, Angular, Node.js, Express.js, YARP	Cloud & DevOps AWS EC2, S3, ELB/ALB, CloudSearch, RDS, WAF, Docker, GitLab CI/CD, Firebase, Vercel, IIS, Linux
Data & Distributed Systems SQL Server, MySQL, Oracle, Query Store, Redis, RabbitMQ, AWS SQS, SignalR, WebSockets	AI, ML & Quality Qdrant, RAG, AI Agents, MCP, HOG, YOLO, DecisionTreeRegressor, Playwright automation